

PEDAMALLA KARTHIK

AI/ML Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI/ML fresher (BCA in Artificial Intelligence) with hands-on experience building and deploying Machine Learning, Deep Learning, Computer Vision, and Generative AI applications, including a forest-fire prediction model achieving 98% accuracy and a defect-detection system achieving 100% recall. Skilled in Python, SQL, Scikit-learn, TensorFlow, PyTorch, OpenCV, and RAG-based systems, with a strong foundation in data preprocessing, EDA, feature engineering, model evaluation, and CI/CD deployment using Docker and Jenkins. Seeking an entry-level AI/ML Engineer or Data Scientist role.

TECHNICAL SKILLS

Languages: Python, SQL

ML/DL Frameworks: Scikit-learn, TensorFlow, PyTorch, XGBoost, Pandas, NumPy

Machine Learning & Deep Learning: Regression, Classification, Feature Engineering, Predictive Modeling, Model Evaluation, CNNs, OpenCV

Generative AI: LLMs, Prompt Engineering, RAG, OpenAI API

Data Analysis & Visualization: Data Cleaning, EDA, Statistics, Power BI, Excel, Matplotlib, Seaborn

Developer Tools & Deployment: Git, GitHub, Flask, Streamlit, MySQL, Docker, Jenkins, Pydantic, Jupyter Notebook, VS Code

Core Competencies: Problem-Solving, Technical Communication, Team Collaboration, Adaptability, Time Management

INTERNSHIP EXPERIENCE

Besant Technologies

Mar 2026 – Present

Data Science and Generative AI Intern

Bangalore, India

- Cleaned, preprocessed, and analyzed multiple real-world datasets using Python, Pandas, NumPy, Matplotlib, and Seaborn, translating raw data into actionable insights through statistical analysis and visualization.
- Built and evaluated multiple machine learning models with Scikit-learn, applying feature engineering and model-evaluation techniques while communicating findings within a Generative AI project team.

The Skybrisk

Feb 2026 – Aug 2026

Python Developer Intern

Remote

- Developed and debugged Python applications and scripts, applying core programming and object-oriented concepts to solve real-world coding tasks.
- Completed hands-on Python development projects, gaining practical exposure to coding best practices, debugging, and software development workflows.

PROJECTS

AI-Powered Manufacturing Defect Detection System | Python, Deep Learning, Computer Vision, TensorFlow/PyTorch, OpenCV, Streamlit

- Built an end-to-end computer vision pipeline to automatically identify surface and manufacturing defects from product images.
- Achieved 100% recall for defective parts, minimizing false negatives and improving defect detection reliability through a real-time inspection workflow with image preprocessing and model inference.

AI-Powered Customer Churn & LTV Optimization Engine | Python, Scikit-learn, Pandas, Streamlit, Pydantic

- Developed an ML system to predict customer churn and estimate Customer Lifetime Value (LTV) using customer behavior data.
- Built a Streamlit Operations Control Center with validated customer-level inputs, risk predictions, and retention insights to prioritize high-risk customers.

Echo Mind – Multimodal AI Operating System | Python, Streamlit, RAG, OpenAI API, MySQL

- Engineered a multimodal AI assistant integrating document processing, voice interaction, web-based retrieval, and persistent conversation memory.
- Implemented RAG-based contextual retrieval across multiple document and image formats, with MySQL-backed memory to maintain context across user interactions.

Algerian Forest Fire Prediction System | Python, Flask, Scikit-learn, Docker, Jenkins

- Trained a machine learning classification model using weather and environmental features, achieving 98% prediction accuracy for forest fire risk.
- Deployed the model as a Flask application and containerized/operationalized it using Docker and Jenkins to demonstrate CI/CD deployment fundamentals.

EDUCATION

Bachelor of Computer Applications – Artificial Intelligence, Krishna Chaitanya Degree College (85%)

2023 – 2026

Intermediate, Sri Vivekananda Junior College (75%)

2021 – 2023

Secondary School Certificate (SSC), ZPP High School (95%)

2020 – 2021

CERTIFICATIONS

- Besant Technologies – Data Science with Generative AI
- Business Writing Certification – Professional Communication Skills